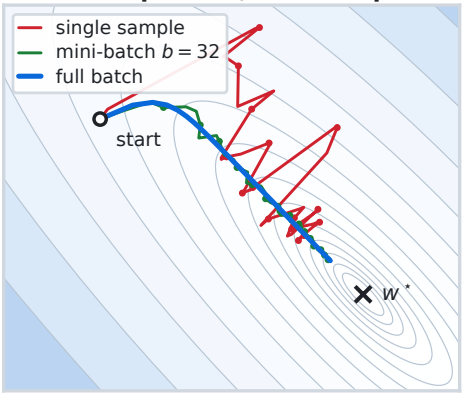
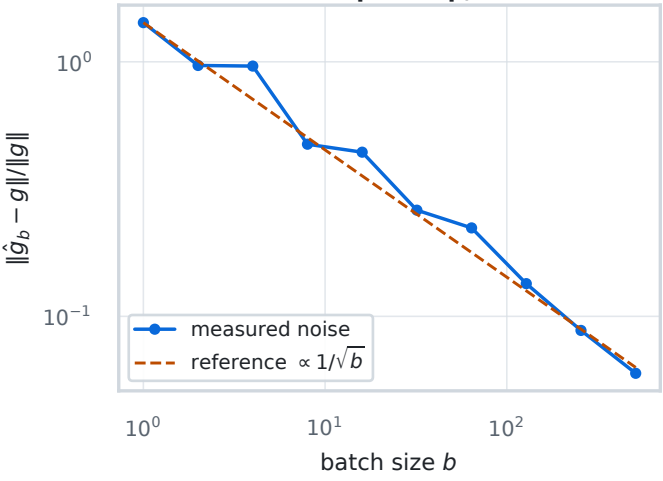

(a) 45 steps each, same step size
single sample
mini-batch $b = 32$
full batch
start
$w^\star$


(b) more data per step, less noise
$\|\hat{g}_b - g\|/\|g\|$
10^0
10^{-1}
measured noise
reference $\propto 1/\sqrt{b}$
10^0
10^1
10^2
batch size b